IEEE TUNISIAN STUDENT AND YOUNG PROFESSIONAL CONGRESS

NEXUS: Next-Generation Emergency eXchange for Universal Sustainability

TSYP14 TECHNICAL CHALLENGE

SCOPE & TOPIC

Tunisia faces increasing climate and infrastructure risks, including Flash floods, Wildfires, Drought and dam shortfalls, Coastal erosion, Summer heatwaves...
The challenge focuses on building low-cost, low-power smart systems that connect sensing, communication and response into one resilient system.
Teams should apply systems thinking to build solutions that detect a crisis early and deliver actionable information to the people who can respond.
Choose one application area:

- Environmental & Climate Sensing — flood, wildfire, drought or air/water-quality monitoring
- Disaster Response Coordination — search-and-rescue, logistics and first-responder coordination
- Critical Infrastructure Resilience — water networks, dams, power grids or buildings
- Public Health & Emergency Monitoring — heatwave, epidemic or air-quality alerts
- Community-Deployable Resilience Systems — solar-powered, offline-capable and locally maintainable systems

Solutions are strongly encouraged to address a real Tunisian region, hazard or community and at least one UN Sustainable Development Goal (SDG).

Challenge problem

Disaster losses are often not caused by a lack of data, but by data that does not reach an early-warning system, responder or household in time to act.
Teams must design a smart solution that:
Detects → Communicates → Coordinates → Enables Response
The solution should be designed for real conditions, including intermittent power, limited connectivity, local maintenance capacity and affordability.

Challenge goal

1. **Sense Before It Escalates**

Use smart sensing to detect hazards such as:
Flood • Fire • Drought • Structural Stress • Heat • Pollution
early enough for meaningful action.

2. **Design for Tunisia's Reality**

Build for:

- Intermittent power
- Intermittent connectivity
- Local maintenance
- Affordability
- Real field conditions

3. **Connect Sensing to Action**

Ensure information reaches the appropriate:
Household • Municipality • Civil Protection • Farmer
in a usable form and in time to act.

4. **Engineer the Whole System**

Apply:

- Systems architecture
- Systems thinking
- Risk management
- Lifecycle resilience

5. **Advance the SDGs**

Explicitly connect the solution to relevant UN Sustainable Development Goals and demonstrate measurable community benefit.

Challenge instructor

Phase1: IDEATION & REPORT EVALUATION
Deadline: 12 October 2026
Required

- Project Report: PDF, maximum 30 pages
- Optional Video Pitch: maximum 2 minutes

Any Student Branch missing the deadline is eliminated from the challenge.

The report must demonstrate:

Clarity & Structure

- Clear organization
- Concise language
- Logical flow

Managerial Aspects

- Realistic implementation timeline
- Defined team roles
- Mission, Vision & Values
- Risk mitigation
- Market analysis
- Potential partners
- Budget estimates
- ROI, profit and cost analysis

The project should present a business-oriented idea supported by financial and market analysis.

Technical Soundness

- Innovation in system architecture
- Technical feasibility
- Tools and methodologies used

Humanitarian Impact

- Real community needs
- Inclusion
- Empowerment of local actors
- Relevant SDGs

Ethical Values

- Equity
- Transparency
- Sustainability
- Community ownership

Video Guidelines

Maximum 2 minutes

- Do not show faces.
- Do not show any visible indication of the school.
- Do not show children's faces if surveys involving children were conducted.
- Voice-over narration is allowed.
- Do not mention names or the IEEE Student Branch.

The video should present:

Problem → Solution & Technical Approach → Community Impact

and include:

Project Name • Mission • Vision • Values

Phase 2: Final Presentation & Prototype

Deadline: 10 December 2026

Required

- Final live presentation
- Working hardware / embedded prototype, simulation or MVP
- A genuine physical sensing or communication component is expected
- GitHub repository
- Recorded or live demo
- Technical paper or abstract — optional
- Circuit/system diagrams
- Sensor and communication architecture
- Project & entrepreneurship methodologies

Pitching

English	5 min presentation
2 min demo	3 min Q&A

Scoring

TOTAL SCORE: 120 POINTS

Initial Phase: Report & Video Evaluation: 55 points

- **Report Quality: 5 pts**
  - Clear structure: 2 pts
  - Concise language: 1 pt
  - Logical flow: 2 pts
- **Managerial Aspects: 10 pts**
  - Realistic implementation timeline: 2 pts
  - Risk mitigation: 2 pts
  - Defined team roles: 2 pts
  - Market analysis: 2 pts
  - Financial analysis: 2pts
- **Technical Soundness: 15 pts**
  - Innovation & efficiency: 3 pts
  - System architecture: 3 pts
  - Ethics & technology: 3 pts
  - Technical feasibility: 3 pts
  - Use of right methodologies / tools: 3 pts

- Humanitarian Impact: 10 pts
  - Addresses real community needs: 2 pts
  - Demonstrates real impact: 2 pts
  - Promotes inclusion: 2 pts
  - SDG alignment: 2 pts
  - Empowers local actors: 2 pts
- Ethical & Values: 10 pts
  - Demonstrates equity: 2 pts
  - Transparency: 2 pts
  - Sustainability: 2 pts
  - Wellbeing: 2 pts
  - Community ownership: 2 pts
- Video: 5 pts
  - Not over 2 minutes: 1 pt
  - Problem: 1 pt
  - Solution & impact: 1 pt
  - Creativity in editing: 2 pts

Final Phase

- Presentation: 8 pts
- Prototype Demonstration: 15 pts
- Feasibility & Real-World Deployment: 10 pts
- Sustainability Demonstration: 7 pts
- GitHub Repository: 5 pts
- Community & SDG Impact Narrative: 5 pts

Final Phase Total: 50 pts

BONUS: up to 15 pts

- Video Pitch Quality & Storytelling: up to 3 pts
- Multi-SDG Alignment (2+ UN SDGs clearly advanced): 2 pts
- Tunisian Community / Municipality Partnership (documented collaboration): 3 pts
- Attending Related Training Sessions organized by the SIGHT/CEDA /SYS/Tech4Good (provide screenshot in the form submission): 1 pt for each training (maximum 5 pts)
- Young Professional (YP, age < 35) on the team: 1 pt
- At least one woman on the team: 1 pt

RULES & CRITERIA

Open to Student Branches (SBs) only

Each SB is represented by one team

Team size: **3–5 members**

Team must include at least:

- 1 IEEE SIGHT member
- 1 IEEE Systems Council member
- 1 IEEE CEDA member

All submitted deliverables must remain anonymous

Solutions must include a genuine smart component, such as:

- Sensing
- Edge processing
- Low-power communication
- AI

Pure software/app-only submissions are not accepted.

Important Dates

- Challenge Infosession: **12/09/2026**
- Phase 1 Submission Deadline: **12/10/2026**
- Final Submission Deadline: **10/12/2026**

Winners

3 WINNING TEAMS

- **1st Place:** 500 TND + Certificate
- **2nd Place:** 300 TND + Certificate
- **3rd Place:** 100 TND + Certificate

Submission

Submit your project and show us what you've built.

Contact



Please contact us through the following email address amira.ouerfelli@ieee.org or zied.kechaou@ieee.org or rahma.fourati@ieee.org for any inquires or clarifications.